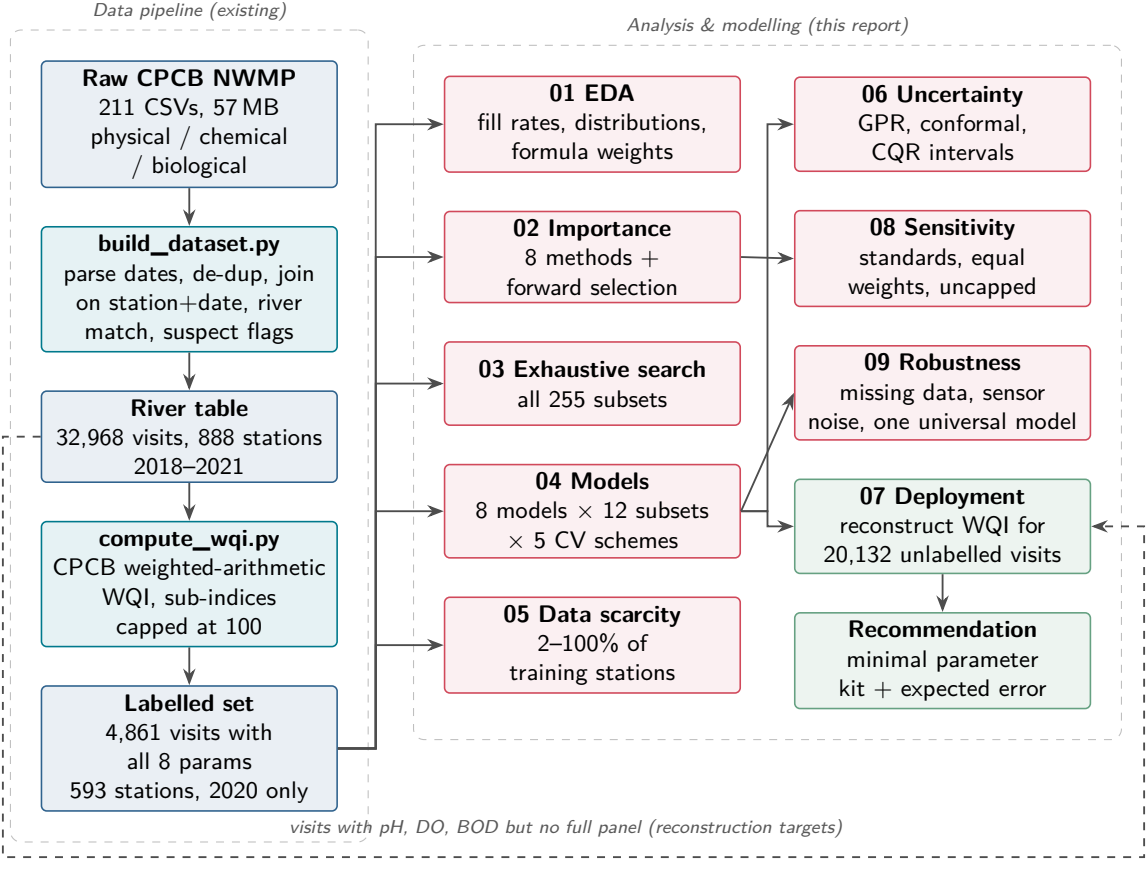
Data pipeline (existing)
Raw CPCB NWMP
211 CSVs, 57 MB
physical / chemical
/ biological
build_dataset.py
parse dates, de-dup, join
on station+date, river
match, suspect flags
River table
32,968 visits, 888 stations
2018–2021
compute_wqi.py
CPCB weighted-arithmetic
WQI, sub-indices
capped at 100
Labelled set
4,861 visits with
all 8 params
593 stations, 2020 only
Analysis & modelling (this report)
01 EDA
fill rates, distributions,
formula weights
02 Importance
8 methods +
forward selection
03 Exhaustive search
all 255 subsets
04 Models
8 models × 12 subsets
× 5 CV schemes
05 Data scarcity
2–100% of
training stations
06 Uncertainty
GPR, conformal,
CQR intervals
08 Sensitivity
standards, equal
weights, uncapped
09 Robustness
missing data, sensor
noise, one universal model
07 Deployment
reconstruct WQI for
20,132 unlabelled visits
Recommendation
minimal parameter
kit + expected error
visits with pH, DO, BOD but no full panel (reconstruction targets)